Haig-Simons Income*

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Abstract

TBD

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1 Introduction

The year 2021 yielded \$16.2 trillion in aggregate capital gains on assets held by households in the United States — 94% of net national income, and an amount larger than wages, dividends, or interest income. While 2021 returns were unusually large, economically signifcant gains have become increasingly common: over the past two decades, real asset appreciation averaged 20% of national income. The rise of capital gains presents challenges for both macroeconomic accounting, where capital gains are not included in national income, and taxation, where capital gains are uniquely subject to a realization rule (they are taxed only when assets are sold). Capital gains are particularly prevelant amoung the wealthy, and have been the subject of numerous recent policy proposals.[1]

Recent scholarship has explained the rise in asset values by secular changes in earnings and profits (**?**), driven by monopoly rents (**??**), retained earnings, intangible investment (**?**), a shift in income to the corporate sector (**?**), and a rise in free-cash flow relative to earnings (**?**). The rise in capital gains have in turn been linked to widespread macroeconomic consequences: higher spending and local employment in response to stock market returns (**?**), greater household wealth (**?**), and elevated leverage prior to the Great Recession (**?**). Less is known, however, on *who* receives the gains— which shapes not only the strength of such aggregate effects but also directly affects measures of the income and wealth distribution.

The primary goal of this paper is to estimate the distribution of capital gains by asset class at the individual level, and quantify their relationship with income, wealth, realized capital gains, and demographic groups. We then apply our new estimates to three key policy-relevant questions: how capital gains affect measures of income inequality, the effective tax rate on capital gains and comprehensive income, and the impact of gains on wealth inequality.

We do so using internal tax data from 2002-2021. The advantage of tax data is its comprehensiveness, spanning the entire income and wealth distribution. The disadvantage is that only realized sales of assets directly show up on tax forms, which are only a small fraction of total appreciation (the sum of unrealized and realized). Prior research has found only about 1/3rd of the stock of capital gains have been realized.[2]

To estimate total (unrealized and realized) gains at the individual level, we use a three step procedure: (i) link individuals to their specific portfolio holdings (ii) capitalize income to estimate wealth (iii) estimate capital appreciation by multiplying wealth by asset class specific returns. In doing so, we exploit never before used tax forms, new data sources, and improved estimation techniques to provide the most accurate available measures of wealth and capital gains at the

[1] These include recent Republican proposals to exempt owner-occupied housing from gains and index gains to inflation, as well as Democratic proposals to tax unrealized capital gains and eliminated the step-up basis at death.

[2] See **?** for Norway, **?** and **?** for the United States.

individual level. For all asset classes we capitalize wealth with heterogeneous cap rates and estimate gains with heterogeneous returns. We highlight four innovations here: (i) we use techniques from the network economic literature to better track partnership wealth through network chains of ownership, which allows us to trace 90% of partnership assets to their ultimate owners (ii) we use never before used tax forms to link 100% of rental real estate assets to their ultimate owner, which allows for a more accurate estimate of real estate wealth and capital gains (iii) we use a new data source on private business sales to better estimate business valuations and capital gains (iv) we use heterogeneous dividend yields and stock market betas from a large sample of discount brokerages to estimate public equity capital gains.

We document several new empirical facts. First, comparing taxable realized gains and aggregate revaluations, we find less than 20% of total gains were subject to taxation. This low rate is driven primarily by individuals delaying realization, as well as a growing share of assets held in tax exempt accounts. The correlation of capital gains and tax realizations is only XXX. Demographically, capital gains concentrated amongst the elderly, with gains rising linearly in age. As the population ages across our sample period, so does the concentration of capital gains in the aged. Amongst individual filers, men have substantially higher gains than women. Across geography, [states that have high KGs]. Their correlation with labor income, capital income, wealth, etc.

The distribution of capital gains is right-skewed, with higher concentration than any other form of income: the top 1% of the distribution received 45% of gains over the sample period. They are concentrated amongst the upper tails of the income and wealth distribution: the top 1% (10%) received 45.3% (75.7%) of total revaluations over our sample period. The wealthiest 1% received 90% of capital gains, while holding only 60% of wealth.

We estimate the distribution of a comprehensive measure of Haig-Simons income that is inclusive of capital gains. Because capital gains are highly concentrated, they increase measured income inequality. The top 1% (10%) of individuals accrued 21.0% (47.9%) of Haig-Simons income; their share without capital gains is 18% (45%). Capital gains on public and private equity are the main driver of the increased concentration, while housing and pension capital gains lead to a more equal distribution of income.

We estimate the effective tax rate on aggregate capital gains and Haig-Simons income. Althought the average marginal statutory tax rate on realized gains was 18%, due to low realization rates the the average effective nominal tax rate on total gains was just 3% and 5% for real gains. Low tax rates on capital gains in turn lower the tax rate on Haig-Simons income. Surprisingly, high income and wealth groups realize gains at a someone higher rate, and this combined with higher statutory rates lead to higher tax rates on total capital gains. However, since high income and wealth groups have a larger share of gains, their Haig-Simons tax rate is pulled down. As a result, the tax-system is substantially less progressive when measured by Haig-Simons income, with tax rates roughly flat across the income distribution.

Our results on the effects of income concentration and tax progressivity are robust to a variety of alternative assumptions about the allocation of labor and capital income, as well as the incidence of taxes and government spending. While the *level* of income concentration depends on whether we distribute non-capital gain income according to **?** or **?**, capital gains always substantially raises measured income inequality. Our results are also robust to a variety of alternative measurement assumptions for capitalization rates, returns, and unmeasured income.

Finally, we show that the secular increases in stock prices, business values, and housing prices substantially increased the welfare of asset holders over our time period. A long-standing criticism of Haig-Simons income is that when asset price changes are caused by declining interest rates and/or changes in discount rates, individual welfare cannot be computed from the change in asset prices, since in addition to the endowment effect of higher asset values there is an offsetting effect of future lower returns. In a series of robustness exercises, we show, consistent with prior research, the majority of long-run capital gains over the sample period were caused by changes in dividend flows rather than interest rate changes and/or discount rate changes. In addition, we construct capital gains series purged of interest rate and discount rate variation, and show that the resulting distributional series are largely unchanged.

2 Realizations

Capital gains are a relatively widespread form of income, with 80% of the sample having a non-zero gain. Realizations, however, are relatively rare: 83% of individuals report exactly zero gains, and an additional 7% reporting capital losses (of which 4.2% are the maximum deductible amount). The correlation of total and realized capital gains at the individual level is .40, which is surprisingly high given the lumpiness of realizations and the measurement error for total gains. After aggregation across individuals to smooth realization and measurement variation, the correlaiton is even higher: by age the correlation is .7, by income decile .8.

3 Theoretical Framework

Before measuring capital gains and Haig-Simons income, we formally define them and explore how they are related to traditional national accounting concepts.

Output Y_t is produced by final goods firms with capital K_t and labor L_t with production function $Y_t = F(K_t, L_t)$. Firms produce output by hiring labor and capital, making profits of $\Pi_t = Y_t - w_t L_t - \rho_t K_t$. All capital is owned by the business sector, but individual firms rent the capital used in their own production from other firms.

Firms make revenue from profits Π_t and renting capital, $\rho_t K_t$. Out of this revenue, they pay a dividend D_t, and retain earnings RE_t, which which they use to invest in the capital stock.

The capital stock evolves according to $K_{t+1} = (1-\delta)K_t + I_t$. Investment equals retained earnings (corporate savings) plus personal saving, $S_t^p = w_t L_t + D_t - C_t$. Personal saving is invested in firms through new equity issuances EQ_t.

- GDP Y_t equals National Income, which equal the sum of wages, rental income, and profits: $NI_t = w_t L_t + \rho_t K_t + \Pi_t$.
- Business income, defined as pure profits plus rental income is either paid out as dividends or retained within the firm: $\rho_t K_t + \Pi_t = RE_t + D_t$.
- Output equals consumption plus investment, and investment equals the sum of personal and corporate savings: $Y_t = C_t + I_t = C_t + RE_t + S_t$.

The market value of firms V_t equals the value of the capital stock, K_t, plus the market value of the present discounted value of pure profits: $V_t = K_t + E_t \sum_{s=0}^{infty} \Pi_{t+s} D_{t+s}$, where D_{t+s} is the stochastic discount rate.

Capital gains are defined as the change in market value of firms minus net equity issuances: $V_{t+1} - V_t - EQ_t$. They can be a function of:

- A change in the expected value of future profits Π_{t+s}
- Retained earnings RE_t reinvested in the firm
- Changes in the stochastic discount rate D_{t+s}

As defined, capital gains include retained earnings, which are included in national income. We therefore define pure capital gains $KG_t^{pure} = V_{t+1} - V_t - RE_t$.

4 Data & Methodology

Our primary data source is internal individual and business tax records from 2002-2021, provided by the Internal Revenue Service through the Joint Statistical Research Program. We use a 0.2% weighted sample of individual returns (approximately 300,000 observations per year) and 100% samples of business and rental property returns. We supplement the tax data with aggregate series from the Financial Accounts, NIPA, and the Survey of Consumer Finances (SCF). Table 1 presents summary statistics.Table 1 presents summary statistics.

The starting point for our analysis is the Distributional National Accounts of PSZ, which we replicate with internal tax records. We add to this our own estimates of individual level capital gains, estimated via a three-step procedure: (i) link individuals to their specifc portfolio holdings (ii) estimate the market value of the portfolio through capitalization income flows (iii) multiply portfolio value by price appreciation to derive capital gains.

Our capitalization procedure follows PSZ and SZZ: observed tax return income flows are used to estimate asset values by multiplying the flows by a capitalization factor. While this is noisy at the individual level, when averaged at the group level (for example, the top 1% income share, the top 1% wealth share) average wealth will be unbiased as long as, conditional on the group, our estimate of the average capitalization factor is unbiased. A key difference in this paper is we allow for greater heterogeneity in capitalization factors, and thus can account for finer differences in cap factors across income and wealth groups. Table YYY summarizes the capitalization factors used we use for each asset class. For our estimate of capital gains to be unbiased, the same condition for returns must hold: for each income and wealth group, our estimate of average return must be unbiased. Table ZZZ, column B, summarizes the returns used for each asset class.

We now summarize our estimation procedure for each asset class, with details provided in Appendix J.

4.1 Public equities

Prior work capitalizes dividends and/or realized capital gains with a homogeneous cap factor. We use account-level brokerage data a large discount brokerage, provided by Barber & Odean (2000), which contains trading records and portfolio holdings for approximately 78,000 households. We use the data to estimate heterogeneous dividend yields portfolio betas by 25 cells of age and income. Dividend yields increase linearly in age, with the 65+ group averaging 20% higher yields than the under-35 group; conditional on age, they are weakly correlated with income.

Because the brokerage data spans 1991–1996, we estimate yields *relative* to the market average and apply these ratios to current-year aggregate yields. Individual public equity wealth is then estimated as nonqualified dividends divided by the group-specific yield. Table NN compares distributional statistics from our heterogeneous yields with the homogeneous-yield approaches of PSZ and SZZ.

For capital gain yields, we estimate cell-level portfolio alphas and betas from a CAPM-style regression of portfolio returns on the market, allowing returns to differ by both average excess return and market exposure. We then scale aggregate public equity wealth to match the Financial Accounts.

4.2 Private business wealth

Valuation. We estimate firm-level enterprise values by capitalizing EBITDA and sales using valuation ratios derived from two large databases of private business transaction data, the first use of these data for distributional wealth estimation. We estimate capitalization rates by year, two-digit NAICS industry, firm size, and legal form; final enterprise value is the average of sales-based and EBITDA-based valuations. Net financial assets from Schedule L are added to convert enterprise values to market values. Details on the BVR data and capitalization rate estimation are in Appendix J and a companion paper (**?**).

Aggregate results. Our estimates yield \$17T in aggregate private business wealth in 2017, more than \$6T above SZZ and PSZ, who scale to the Financial Accounts. The gap is largest for partnerships (\$8T vs. \$4T), because the FA values partnerships at book value and does not apply market multiples. We argue that the FA likely understates partnership wealth, as intangible capital has risen (**??**) and pass-through activity has expanded (**??**). Concentration is also substantially higher in our estimates, driven by heterogeneous capitalization rates that value larger businesses at higher multiples and by the high concentration of indirectly owned partnership wealth.

Ownership allocation. S-corp and sole proprietorship wealth is allocated directly from Schedule K-1 and Schedule C returns. Partnership allocation is more complex because partnerships can be owned by other partnerships, trusts, or corporations. Prior work has traced business ownership solely through direct holdings reported on Form 1065 K1.

To allocate this wealth to individuals, we apply a result from the network economics literature. Let C denote the partnership ownership matrix, where C_{ij} is the fraction of partnership j owned by partnership i, and let a be the vector of fundamental asset values estimated above. The total market value vector satisfies $V = a + C \cdot V$, yielding:

$$V = (I - C)^{-1} \cdot a \tag{1}$$

We estimate C from the universe of Form 1065-K1 returns. Of \$7T in partnership enterprise value in 2017, \$6T can be traced to a K1 owner, of which \$2.7T flows to individuals; the remainder is held by C-corporations, trusts, foreign owners, and 10% by unidentified entities.

Table K compares the share of private business wealth by the top 10% and top 1% using our valuation method, and compares it to SZZ and PSZ. Concentration is substantially higher in our estimates as compared to prior work. This is a function of (i) heterogeous capitalization rates, values richer individual's businesses, which are larger, at higher multiples (ii) the fact that indirectly owned businesses are much more concentrated than directly owned businesses.

4.3 Tenant occupied housing

Both PSZ and SZZ allocate the total value of tenant occupied real estate, *including real estate owned by partnerships and S-corps* by capitalizing rental income from Part I of Schedule E *from properties that individuals directly own*, i.e. not including rental income from indirectly owned properties. This procedure is potentially problematic, because by capitalizing only Part I Schedule E income only they impute all the value of partnership and S-corp owned housing to the individuals that directly own properties, and none to the individuals that actually own and control this housing.

In this paper we capitalize property.tax payments both directly owned by individuals and indirectly owned through partnerships and S-corps, using data from Form 8825. These data contain property level tax payments, the location of the real estate, and the type of property, which we separate into three broad categories: single family, multifamily, and commercial. We estimate real estate values by capitalizing property tax payments using average county-level property tax rates by property type. Data on effective tax rates is taken from the Lincoln Institute of Land Policy and the Minnesota Center for Fiscal Excellence.

Table NN displays results for 2021. About *half* of t.o. housing is owned by partnerships, which implies that prior estimates were incorrectly allocating a substantial fraction of wealth: the correlaiton of our housing wealth measure and the PSZ/SZZ version is only .45. In addition, our measure is substantially more concentrated than PSZ, since indirectly owned real-estate is concentrated in the upper percentiles of the distribution.

4.4 Owner occupied housing

Following **?**, we capitalize property tax deductions from Schedule A using county-level effective tax rates from the ACS and Decennial Census. This procedure captures approximately 80% of Financial Accounts housing wealth; following PSZ, we allocate the residual to non-itemizers using SCF averages by income decile and age-by-marital-status cell. Capital gains are computed using 5-digit zip code level housing price indices from the FHFA.

The TCJA of 2018 reduced itemization rates from 30% to 11%, dropping our coverage from 77% to 40% of FA housing wealth. To recover non-itemizers' housing values post-2018, we match current-year addresses to 2017 returns where property taxes were itemized and update values using FHFA house price indices, raising coverage back to 75%. The remaining 25% is allocated via SCF averages from post-TCJA survey years.

4.5 Other asset classes

To estimate taxable bond wealth, we use data from SZZ on the average fxed income yield by quantile in the wealth distribution (by year) to capitalize taxable interest fows: 0-99th percentile, 99th-99.9th, 99.9-99.99, and top 01%. For indirectly held bonds held through mutual funds, money-market funds, ETFs, and closed-end funds, we capitalize non-qualifed dividends under an equal returns assumption using Financial Accounts totals. For munis, we capitalize tax-exempt interest. For currency, we follow PSZ and allocate Financial Account wealth by income rank using SCF data.

Table 1: Summary of Wealth and Capital Gains Estimation Methods

Asset Class	Wealth Capitalization	Capital Gains Return
Public equity	Het. yields (Barber & Odean (2000))	Het. portfolio α and β (Barber & Odean (2000))
Private equity	Het. EV/EBITDA and EV/Sales (DealStats)	Het. returns calculated as change in EV minus investment
Owner-occupied housing	Het. property tax rates (ACS/Census)	Het. returns (FHFA price indices)
Tenant-occupied housing	Het. property tax rates (Lincoln Institute)	Het. returns (Freddie Mac and CoStar (commercial))
Fixed income	Het. yields (SZZ)	Excluded
Pension	Capitalize 60% taxable pensions, 30% wages, 10% tax-exempt pensions (PSZ)	Homogeneous (Financial Accounts)

Notes: Table summarizes the capitalization method used to estimate wealth and the return method used to estimate capital gains for each asset class. "Homogeneous return" denotes that aggregate Financial Accounts capital gains are divided by aggregate wealth to form a common yield applied to all individuals. "Heterogeneous" denotes that capitalization factors or returns vary across individuals based on asset or location characteristics. SZZ refers to **?**; PSZ refers to **?**. Fixed income capital gains are excluded from the analysis. See text for details.

5 Demographic and Geographic Concentration

Capital gains are concentrated among older individuals, reflecting the cumulative nature of asset accumulation over the life cycle and the nature of gains to compound due to low realization rates. Figure **??** displays average capital gains by age, which rise steadily from young adulthood through retirement. The age-gains profile is approximately linear through working ages before leveling off after age 70, and fairly tracks the estimated wealth profile. Notably, capital gains do not decline substantially among the oldest age groups, consistent with the well-documented failure of wealth to decumulate in retirement as predicted by standard life-cycle models (**?**). Bequest motives, precautionary saving against health and longevity risk, and the concentration of wealth among households with low marginal propensities to consume all contribute to the persistence of wealth—and hence capital gains—into advanced ages.

Figure **??** also displays capital gain realizations, scaled as a percentage of total gains for that age group. Realization rates are low throughout adulthood and rise slowly until retirement, where they plateau at 65% of total gains. Notably, realization rates never exceed total gains, consistent with evidence on high savings rates out of capital gains and low marginal propensities to consume among wealthy households (**???**). Table **??** summarizes the distribution of capital gains across age groups. Given the strong age-relationship documented above, capital gains are concentrated amongst retirees: individuals aged 65 and over received 36% of total capital gains while comprising only 18% of the adult population. The aging of the U.S. population has contributed to increased concentration of capital gains among the elderly over our sample period. Between 2002 and 2021, the share of capital gains received by individuals aged 65 and over rose from 23% to 36%.

graphs/kg_age_profile.png

Figure 1: Average Capital Gains by Age

Notes: Figure displays average annual capital gains by 5-year age bracket, pooled over 2002–2021. Capital gains include unrealized and realized appreciation on equities, housing, pensions, and private businesses. Shaded area indicates 95% confidence interval. Sample restricted to tax units with positive wealth.

Table 2: Distribution of Capital Gains by Age Group

Age Group	Mean KG ($)	Share of Gains	Share of Gains by Asset: Stock	Business	Housing	Share of Pop.
20–44	[XX]	[XX]%	[XX]%	[XX]%	[XX]%	[XX]%
45–64	[XX]	[XX]%	[XX]%	[XX]%	[XX]%	[XX]%
65+	[XX]	[XX]%	[XX]%	[XX]%	[XX]%	[XX]%
All	[XX]	100%	100%	100%	100%	100%

Notes: Distribution of capital gains by age group, 2002–2021. Stock includes public equity and mutual funds. Business includes partnerships, S-corps, and private C-corps. Housing includes owner-occupied and rental real estate.

Capital gains are geographically concentrated in states with concentrations of financial assets appreciating housing markets. Figure **??** displays capital gains per capita by state, averaged over 2002–2021. The geographic distribution is highly uneven: residents of the District of Columbia received an average of $[19,500] in annual capital gains – more than double the national average – while residents of West Virginia received just $3,500. The top five states—the U.S. Virgin Islands, the District of Columbia, California, New York, and Florida—combine high concentrations of financial wealth (New York), appreciated housing markets (California, New York, Florida), and favorable tax treatment (the Virgin Islands, Florida). The bottom five states—West Virginia, Mississippi, Indiana, Kentucky, and Alabama—share characteristics of lower household wealth and below average housing returns.

State-level capital gains tax rates and the presence of an estate tax are only moderately correlated with individual-level capital gains, with an important exception for the top .1% of the distribution. Table YY regresses state-level top capital gains tax rates on an individual's rank in the capital gains distribution. Column (1) includes only year fixed effects, while Column (2) adds controls for labor income and age. In both specifications, there is a moderate positive correlation between capital gains and tax rates, suggesting other locational characteristics may dominate the location choices. Column (3) adds dummy variables for the 99-99.9th and 99.9th-100th percentile of the capital gains distribution. For the top capital gain group, there is an economically and statistically significant negative coefficient: the average state tax rate of this group is .92 percentage points lower, compared to an average top-state capital gains tax rate of 5.29%. Columns (4)-(6) consider the relationship between capital gains and the presence of an estate tax, with the regression restricted to retirees (age 65+). The results show that the very largest capital gain holders (top .01%) are 9.5% less like to reside in a state with an estate tax, compared to a sample mean of 25%.

6 Appendix

6.1 Demographic and Geographic

Table 3: Characteristics of Top Capital Gains Recipients

Percentile Group	Average Age	Average Capital Gains ($)
Top 10%	[XX]	[XX]
Top 1%	[XX]	[XX]
Top 0.1%	[XX]	[XX]
Top 0.01%	[XX]	[XX]

Notes: Table reports average age and capital gains for individuals in each percentile group, pooled over 2002–2021. Percentiles based on annual capital gains.

Table **??** shows the breakdown of gains for married versus unmarried filers, and amongst the unmarried, female versus male. Average capital gains are substantially higher for married filers than for unmarried filers, reflecting large differences in income and wealth. Among unmarried filers, men have moderately higher average capital gains than women, again reflecting differences in wealth: [XX] compared to [YY]. The largest differences in gains between men and women concern private businesses, with average male capital gains double that of females. This pattern reflects the well-documented gender gap in entrepreneurship: men are roughly twice as likely as women to start and own businesses (**?**).

Table 4: Income, Wealth, and Capital Gains by Filing Status and Gender

Group	Income ($)	Wealth ($)	Capital Gains ($)
Married	[XX]	[XX]	[XX]
Unmarried Male	[XX]	[XX]	[XX]
Unmarried Female	[XX]	[XX]	[XX]

Notes: Table reports average income, wealth, and capital gains by filing status and gender, pooled over 2002–2021. Married includes all joint filers. Unmarried includes single, head of household, and married filing separately.

Table 5: Top 5 States for Capital Gains Per Capita by Asset Class

Rank	Overall	Equity	Real Estate	Business
1	Virgin Islands	DC	DC	Wyoming
2	DC	Connecticut	California	American Samoa
3	California	Wyoming	Hawaii	DC
4	New York	New York	Washington	Connecticut
5	Florida	Massachusetts	Florida	New York

Notes: Table reports states and territories with highest capital gains per capita by asset class, pooled over 2002–2021. Equity includes public stocks and mutual funds. Real estate includes owner-occupied and rental property. Business includes partnerships, S-corps, and private C-corps.

Looking at a more granular level, Table **??** presents population-weighted OLS regressions of capital gains per capita on commuting-zone characteristics, with each correlate entered individually (columns 1–8) and jointly (column 9). Several patterns emerge. Capital gains are [higher/lower] in more urbanized commuting zones, with a one-standard-deviation increase in metro share associated with $[XX] higher gains per capita (column 1). The "superstar city" premium (**?**) is particularly pronounced: gains are $[XX] higher per capita in CZs containing superstar cities (column 2), [and this premium remains large after conditioning on metro share in column 9, indicating that it is not urbanization per se but specifically high-cost metros that concentrate gains]. Commuting zones with a larger elderly population share have significantly higher capital gains (column 3), consistent with the life-cycle patterns documented above. Gains are negatively correlated with the **?** causal measure of intergenerational mobility (column 4): commuting zones where children from low-income families have the worst outcomes tend to have the highest capital gains, consistent with gains accruing disproportionately to areas marked by high wealth concentration and residential segregation. Housing supply elasticity (**?**) is [negatively] associated with gains (column 5), consistent with supply constraints capitalizing into asset values [—though this correlation is substantially attenuated once superstar status is controlled for in column 9]. Larger commuting zones have [higher] per capita gains (column 6). The share of Black residents is [positively/negatively] associated with capital gains per capita (column 7), consistent with [gains concentrating in diverse metropolitan areas / the racial wealth gap limiting gains in majority-Black communities]. The share of college-educated adults is [positively] associated with gains (column 8), consistent with higher rates of stock market participation and wealth accumulation among college-educated populations. The joint specification in column 9 explains [XX]% of the cross-CZ variation in gains, with [superstar status/elderly share] as the strongest predictors.

graphs/kg_map_total.png

Figure 2: Capital Gains Per Capita by State

Notes: Figure displays average annual capital gains per capita by state, pooled over 2002–2021. Capital gains include unrealized and realized appreciation on equities, housing, pensions, and private businesses. States grouped into quintiles.

Table 6: Correlates of Capital Gains Per Capita Across Commuting Zones

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Metro share	[XX]								[XX]
	([XX])								([XX])
Superstar city		[XX]							[XX]
		([XX])							([XX])
Share age 65+			[XX]						[XX]
			([XX])						([XX])
Chetty mobility (p25)				[XX]					[XX]
				([XX])					([XX])
Saiz elasticity					[XX]				[XX]
					([XX])				([XX])
log(CZ population)						[XX]			[XX]
						([XX])			([XX])
Share Black							[XX]		[XX]
							([XX])		([XX])
Share college								[XX]	[XX]
								([XX])	([XX])
Observations	[XX]	[XX]	[XX]	[XX]	[XX]	[XX]	[XX]	[XX]	[XX]
R^2	[XX]	[XX]	[XX]	[XX]	[XX]	[XX]	[XX]	[XX]	[XX]

Notes: Table reports population-weighted OLS regressions of capital gains per capita on commuting-zone characteristics, pooled over 2002–2021. Dependent variable is average annual capital gains per capita ($). Metro share is the share of CZ population in metropolitan counties. Superstar city is an indicator for CZs containing a superstar CBSA as defined in **?**. Share age 65+ is the fraction of CZ population aged 65 and over. Chetty mobility (p25) is the causal effect on child income rank at age 26 for parents at the 25th percentile (**?**). Saiz elasticity is the housing supply elasticity from **?**. Share Black is the population-weighted share of Black residents (2019 Census population estimates). Share college is the population-weighted share of adults aged 25+ with a bachelor's degree or higher (ACS 2017–21). Standard errors in parentheses.

The geographic concentration of capital gains has important implications for state tax revenues. Because most states tax capital gains as ordinary income, states with large concentrations of capital gains experience pronounced revenue volatility tied to asset market performance. California's Legislative Analyst's Office estimates that capital gains are nearly three times as volatile as the state's overall personal income tax base, and account for roughly 40% of total income tax volatility (**?**). More broadly, personal income tax revenue volatility has increased across the majority of states in recent years, driven in significant part by greater reliance on increasingly volatile capital gains realizations (**?**). California and New York—the two states with the largest concentrations of capital gains—accounted for the majority of the national decline in state income tax revenues in fiscal year 2023 (**?**).

Beyond revenue volatility, the geographic concentration of capital gains implies that standard measures of regional income divergence understate the true extent of spatial inequality. Most studies of geographic income convergence—including the influential work of **?**—measure income using wages or pre-tax factor income that excludes capital gains. Because capital gains are concentrated in already-prosperous states such as California, New York, and Connecticut, incorporating gains into a comprehensive income measure would substantially widen measured geographic divergence. **?** documents the growing concentration of high-skill, high-wage employment in a small number of "superstar" cities; our data suggest that these same cities are also the primary recipients of capital gains, compounding the divergence in economic fortunes across regions.

The geographic concentration of real estate capital gains has particularly important implications for housing affordability and spatial misallocation. In supply-constrained housing markets, asset appreciation accrues primarily to incumbent homeowners, widening the divide between owners and renters and reinforcing barriers to geographic mobility. **?** estimate that land-use restrictions in high-productivity cities such as New York, San Francisco, and San Jose lowered aggregate U.S. GDP growth by more than a third between 1964 and 2009 by preventing workers from relocating to their most productive locations. Housing capital gains in these markets function as a transfer from future buyers to current owners, raising the entry cost for workers seeking access to high-wage labor markets. Recent work by **?**, however, challenges the supply-constraint explanation, finding that income growth—not housing supply elasticity—is the primary predictor of cross-city house price growth over 1980–2020. If income growth rather than regulation drives geographic price divergence, the omission of capital gains from standard income measures is all the more consequential: capital gains represent a large and geographically concentrated component of comprehensive income that is absent from the income measures used in housing demand models. The geographic pattern in our data—with California, Hawaii, Washington, and the District of Columbia leading in real estate gains per capita—is consistent with both supply-constraint and income-driven explanations of housing price divergence, and underscores the need for further research disentangling these channels.

The spatial distribution of capital gains also intersects with intergenerational mobility and opportunity. **?** document substantial geographic variation in rates of upward mobility across commuting zones, driven in part by local conditions such as school quality, social capital, and residential segregation. Capital gains reinforce these geographic disparities through at least two channels. First, wealth bequests—amplified by the step-up in basis at death, which eliminates the tax on unrealized gains—allow high-gain families to transfer substantial resources to the next generation in a geographically concentrated manner. Second, real estate appreciation affects local public finance: because property taxes fund the majority of K–12 education, areas with rising home values generate more per-pupil revenue, potentially widening educational quality gaps between high-gain and low-gain regions.

The geographic concentration of capital gains may itself be partly endogenous to state tax policy. States without income taxes—including Florida, Texas, Nevada, Wyoming, Tennessee, and South Dakota—attract high-wealth individuals seeking to minimize their tax burden on realized gains. **?** document that state taxes significantly affect the location decisions of "star" scientists and inventors, with a 1 percent increase in the top tax rate reducing the stock of star scientists by 1.6 percent. **?** find that millionaire tax migration, while modest in aggregate, is concentrated among the highest-income households and responsive to cross-state tax differentials. This sorting behavior reinforces the geographic concentration of capital gains: Florida's appearance in the top five states for overall gains is likely driven in part by the in-migration of high-wealth retirees attracted by the absence of state income taxes, rather than solely by local asset appreciation.

Finally, the geographic concentration of capital gains creates heterogeneous local wealth effects that amplify regional business cycle asymmetries. **?** show that declines in housing wealth during the Great Recession had large effects on household consumption, with a marginal propensity to consume out of housing wealth of approximately 5–7 cents per dollar. Because capital gains—and the underlying asset holdings—are geographically concentrated, asset price booms disproportionately stimulate consumption and employment in already-prosperous areas, while busts devastate less-diversified regions. The experience of Nevada and Arizona during the 2008 housing crash illustrates this vulnerability: states with outsized exposure to real estate gains experienced the most severe contractions when prices reversed. The time-series volatility of capital gains at the state level thus represents a significant source of asymmetric regional risk, with implications for both macroeconomic stabilization policy and the design of fiscal federalism.

6.2 Private Business Details

We estimate capilization rates for EBITDA (β_{ct}^{EB}), and sales (β_{ct}^{SA}) by year by averaging transaction multiples by cell, for cells defined by two-digit NAICS industry, firm size (measured by sales), and legal form of organization. Table NN explores how business cap rate by these cell variables by regressing EV/SA and EV/EBITDA ratio on firm size, legal form of organization, and NAICS industry. Firm size is a particularly important predictor of valuation multiples. Being in the top decile of size is associated with an increase in EV/EB of 1.3 and EV/SA of.11 compared with being in the 5th decile.

We value individual businesses by multiplying sales and EBITDA from their tax returns by the two cap rates, and taking an average of the two as the final valuation:

$$\text{Sales EV}_{ft} = \text{Sales}_{ft} \cdot \beta_{ct}^{SA} \tag{2}$$

$$\text{EBITDA EV}_{ft} = \text{EBITDA}_{ft} \cdot \beta_{ct}^{EB} \tag{3}$$

$$\text{Final EV}_{ft} = (\text{Sales EV}_{ft} + \text{EBITDA EV}_{ft})/2. \tag{4}$$

To go from enterprise to market values, we use data from Schedule L of business tax returns to add back net financial assets.

Table NN presents our estimates of aggregate private business wealth, showing a total of \$17 T in 2017, more than \$6 T greater than SZZ and PSZ, both of who scale their totals to the FA. Our valuations for partnerships, \$8T, is substantially larger than the SZZ and FA values (\$4T). The reason for this difference is a difference in methodology: the FA values partnerships based on book value, and do not apply market based multiples.[3]

Individual business wealth equals the value of their underlying businesses times the fraction of the share they own: $W_{it}^{\text{Pr. Bus}} = \sum_f MV_{ft} sh_{ift}$. Allocation of S-corps and soleproprietorship value is straightforward. S-corps can only (with a few exceptions) be held by individuals, and ownership shares are listed directly on Schedule 1120 K1 information returns. Sole proprietorships returns are inherently linked to their owners, because their tax return (Schedule C) is included as part of 1040.

Allocating partnership business wealth is more difficult since partnerships can be owned by not only by individuals, but by other legal forms of organization (such as other partnerships, trusts, or corporations) that ultimately may be controlled by individuals. Prior work has traced business ownership solely through direct holdings reported on Form 1065 K1. In this paper, we will attempt to cut through the chain of ownership and allocate valuation to their ultimate owner.

We do so by applying a theoretical result from the network economics literature. A network of partnership ownership is denoted by matrix C, where $C_{ij} isthefractionofpartnership$j$thatpartnershipiowns.Thevalueofpartnership$i$isthevalueofitsf$ = a˙i + $\sum_j C_{ij} V_j$. The vector describing the market value of all partnerships is then given by V = a + C $\cdot V$. Solving for the total value of the firm, $V = (IC)^{-1} \cdot a$.

We estimate the a vector as described above, by capitalizing business level EBITDA and sales. We estimate the partnership ownership matrix C using Form 1065-K1, which lists the percentage of each partnership that is owned by other partnerships. Table NN provides the results of this analysis. For partnership operating businesses, the aggregate enterprise business value was \$7 T in 2017. Of this, \$6 T can be traced back to some K1 owner, of which ultimately \$2.7 T flows to individual ownership. The remaining values are held by C-corporations, trusts/estates, foreign owners, and 10% by unidentified entities.

Table K compares the share of private business wealth by the top 10% and top 1% using our valuation method, and compares it to SZZ and PSZ. Concen-

[3] We argue that the FA is substantially undervaluing partnership businesses by not applying market multiples. While a simplified book-value approach to valuation was less of a problem when partnerships were a small fraction of aggregate business income, as shown by **?** and **?**, over the past two decades pass through activity has exploded. The rise of pass-through business fundamentally altered the partnership landscape from predominantly small professional business to larger enterprises.[4] In addition, the growing importance of intangible capital – as shown in **?** in a small business context and **?** more generally – has widened the gap between book and market values, as many intangible assets remain unrecorded on balance sheets.

tration is substantially higher in our estimates as compared to prior work. This is a function of (i) heterogeous capitalization rates, values richer individual's businesses, which are larger, at higher multiples (ii) the fact that indirectly owned businesses are much more concentrated than directly owned businesses.

For partnerships and S-corporations, we estimate nominal capital gains between time $t-1$ and t as the change in the market value of the business, minus any change in invested capital. Change in capital is equal to the change in paid in capital plus retained earnings for S-corps, and the change in the capital account for partnerships:

$$KG_{ft}^{Scorp} = \Delta MV_{it} - \Delta\text{Paid in capital}_{ft} - \Delta\text{Retained earnings}_{ft} \quad (5)$$

$$KG_{ft}^{Part} = \Delta EV_{ft} - \Delta\text{Partners Capital Accounts}_{ft} \quad (6)$$

For private C-corporations and sole proprietorships, there is no comparable data for basis changes, and thus we estimate gains using price indexes from public corporations, specifically the S&P 500.

6.3 Public Equity Details

We estimate heterogeneous dividend capitalization rates using a sample of account-level data from a large discount broker, provided by Barber & Odean (2000). Following Chodorow Recih et al and Graham and Kumar (2006), we merge the account level information on holdings with CRSP stock and mutual fund data to calculate individual level dividend yields. We then compute average dividend yields by 25 cells of age and income groups.

Appendix Figure J shows the slope of dividend yield by age and income. Dividend yields increase linearly in age, with aged 65+ showing a distinct preference for income rather than growth stocks: the average yield for this group is 20% larger than the under 35 age group. Splitting by income, the lowest quartile of income in the sample have slightly elevated yields compared to the rest of the distribuion.

The B&O data contains dividend yields from the years 1991-1996. To operationalize these for our sample period of 2002-2021, where average stock yields are somewhat lower, we use the data to estimate dividend yields *relative* to the value-weighted average yield of the market. For each cell c, we estimate relative yield $\rho_c = \bar{d}_c/\bar{d}$, where $\bar{d}_c$ and $\bar{d}$ are, respectiively, the value weighted mean yield for cell c the overall market. Dividend yields in year t and then computed as the cell level as $\bar{d}_{ct} = \rho_c \cdot \bar{d}_t$. Finally, we scale aggregate public equity wealth to match the FA aggregate.

We estimate individual level public equity wealth by dividing nonqualified dividends by group specific dividend yield: $W_{it}^{equ} = div_{it}/\bar{d}_{c(i)t}$. Table NN presents distributional statistics for public equity wealth shares across the income distribution, and compares them with two alternative methods: those of PSZ and SZZ, both of which use homogeneous yields.

In estiimating capital gain yields, we allow for two degrees of return heterogeneity: difference in average excess return (α_c) and market exposure (β_c).

We estimate these parameteres by cell c through a CAPM style regression of portfolio return on the market excess return:

$$R_t^c = \alpha_c + R_t^f + b_t^c(R_t^m - R_t^f) + \epsilon_{ct}. \tag{7}$$

To go from total returns to real capital gain yield γ, the dividend yield and inflation rate are subtracted: $\gamma_{it} = \hat{R}_{c(i)t} - d_{c(i),t} - \pi_t$.